

# Workbook
# PHYSICAL SCIENCES

Name: _____

Class: _____

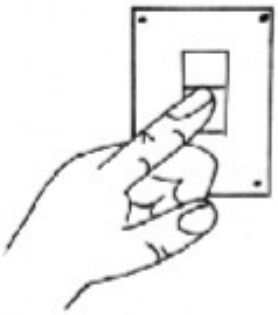
# Forced to Move

FIRST SCHOOL YEARS

## Push or Pull

Name: Date:

- Write whether each picture shows a "push" or a "pull".



_ _ _ _ _



_ _ _ _ _



_ _ _ _ _



_ _ _ _ _



_ _ _ _ _



_ _ _ _ _

- Draw another "push" and "pull" picture.

| push | pull |
|---|---|
| | |

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# Forced to Move

## FORCES AROUND YOU

**Forces** are pushes or pulls. They can start moving, and they can stop, speed up, slow down, or change the direction of moving objects. They can lift things, or cause them to turn, bend, or twist. They can also prevent motion: for example, they can stop a bridge from collapsing.

### ACTIVITY

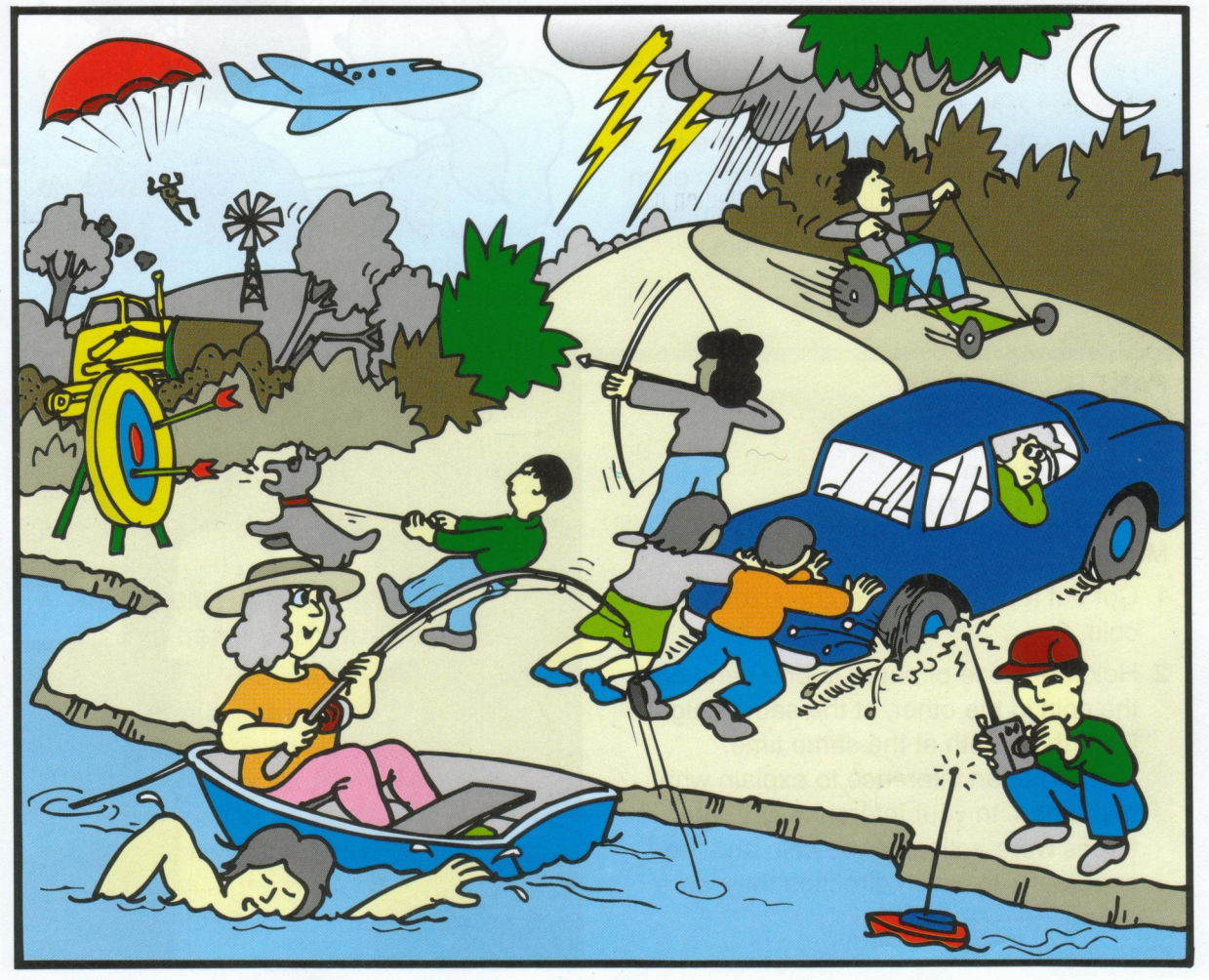
Look at the cartoon below. Can you find examples of:

(a) pushing forces (what is doing the pushing?)

(b) pulling forces (what is doing the pulling?)

Which of the examples are:

- forces caused by gravity?
- lifting forces?
- forces where something bends?
- muscular forces?
- electrical forces?






From the image on the previous page list all forces, decide if they are a push, pull or twist and what type of force is causing the motion.

| Description of force | Push, pull or twist | Type of force |
|---|---|---|
| 1. | | |
| 2. | | |
| 3. | | |
| 4. | | |
| 5. | | |
| 6. | | |
| 7. | | |
| 8. | | |
| 9. | | |
| 10. | | |
| 11. | | |
| 12. | | |
| 13. | | |
| 14. | | |
| 15. | | |
| 16. | | |
| 17. | | |
| 18. | | |
| 19. | | |
| 20. | | |




Forces are pushes or pulls which can change the shape or motion of an object.

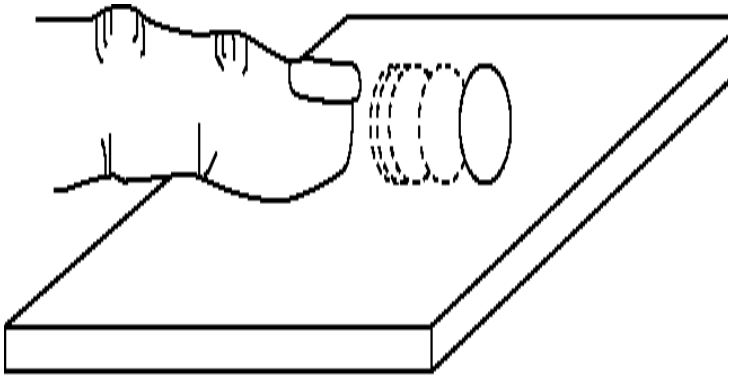
*The only way to describe forces is by describing what they do.*

*Draw up the following table. The first example of an effect a force is given. In class discussion, students give examples of various effects of forces. They complete the table.*

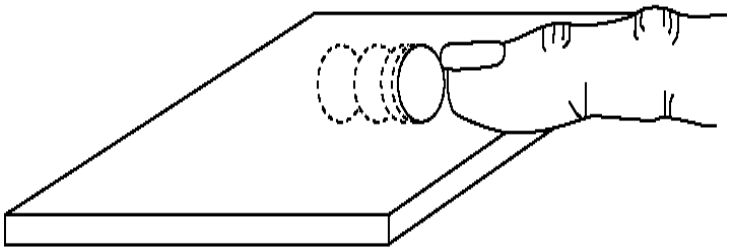
| ***Effect of Force*** | ***Example*** | |
|---|---|---|
| Starting Motion | 1)<br>2)<br>3) | Kicking a stationary ball. |
| Stopping Motion | 1)<br>2)<br>3 | Catching a moving basketball. |
| Speeding up Motion | 1)<br>2)<br>3 | Kicking a ball in the same direction it is moving. |
| Slowing down Motion | 1)<br>2)<br>3 | Space capsule uses a parachute to slow down. |
| Changing Direction of Motion | 1)<br>2)<br>3 | A soccer ball is kicked in the opposite direction it is moving. |
| Changing the Shape of an Object | 1)<br>2)<br>3 | Squeeze toothpaste. |
| Having No Visible Effect | 1)<br>2)<br>3 | A standing building. |

# Forced to Move

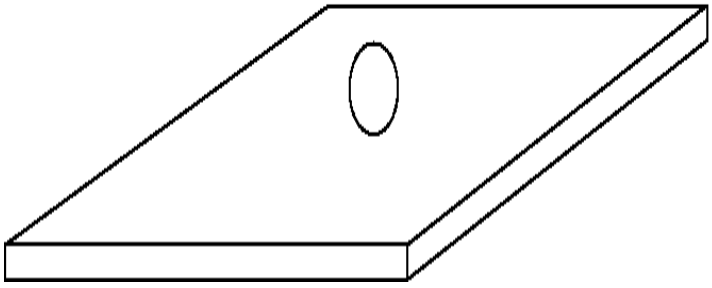
**1** Match the descriptions with the pictures and choose the correct word or words.



When the ball is resting on the table top, the forces are **unbalanced/balanced**.



The ball will move if you push it because the forces are **unbalanced/balanced**. If you stop pushing it, it will **slow down/speed up** because of the friction between it and **your hand/the table top**. If you keep pushing it, it will **slow down/keep moving**. If you push it harder, it will move **slower/faster**.



If you push a rolling ball in the opposite direction to its motion, it will **speed up/slow down**. If you apply a push at an angle to its motion, it might change **direction/speed/direction and speed**.

**2** Choose words from the box to complete the sentences.

# Forced to Move

| energy | friction | average | pull | time | start | speed | stopped |
|---|---|---|---|---|---|---|---|

Force is the general name for a push, a _____ or a twist.

Forces can _____ , stop or change the direction of an object.

Forces give _____ to objects.

_____ is a force that tries to stop things moving.

Motion cannot be started, altered or _____ except by a force.

One way to compare motion is to compare the _____ of objects.

Speed is the distance an object travels in a certain amount of _____ .

When we divide the distance travelled by the time taken, it gives us the _____ speed.

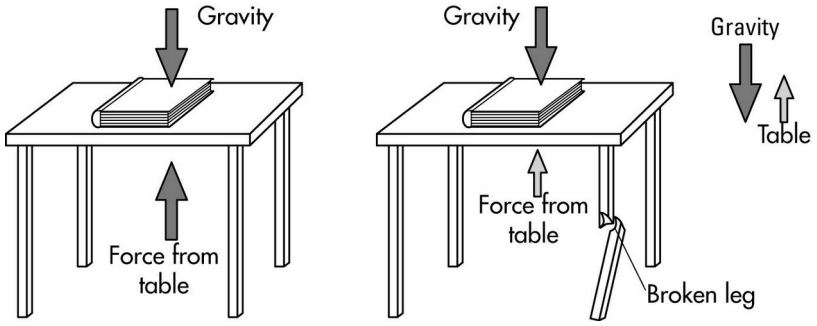
# Forced to Move

**Pushing and pulling forces**

**Key terms**
gravity unbalanced forces

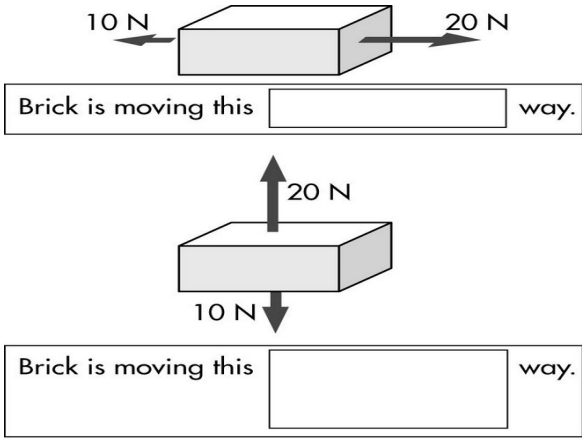
- Forces are pushes or pulls.
- Forces can change the speed of something or change the direction of its movement.
- Force can also change the shape of something.
- Force is measured in units called newtons (N).
- Force can be measured using a spring balance or an electronic force sensor.

**Forces only cause movement when the force is bigger in one direction, when the forces are unbalanced. Look at your books on the table. They are not moving but there are forces acting on them. Gravity is pulling the books downwards and the table is pushing the books back up with the same force. The forces on the books are balancing each other out and the books do not move.**
**If the leg of the table breaks, the force of gravity will pull the books downwards. The books move down because the force pulling down is bigger than the force of the table pushing upwards. The two forces combined result in an overall downward force. This force is called the net force.**




No net force Net force downwards

Label the net forces below and draw an arrow that shows in which direction the object is moving.



# Forced to Move

Rule a line to match each statement with the correct term. (Start the lines from the dots.)

The letter that the line passes through gives the answer to the question at the bottom of the page.

**1** The movement of an object.

**2** The measure of the force of gravity on an object.

**3** The distance an object travels divided by the time.

**4** The unit for measuring force.

**5** The upward force on an object due to the water it has displaced.

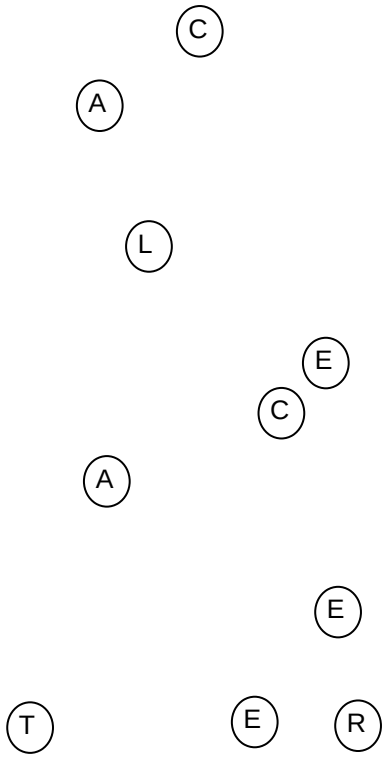
**6** The amount of material in an object.

**7** The rubbing force that opposes motion.

**8** Motion is started, altered or stopped when forces are ___.

**9** The unit for measuring mass.

**10** The force of attraction between the Earth and an object.




weight

upthrust

unbalanced

motion

newton

speed

gravity

kilogram

friction

mass

Pedalling your bicycle harder causes it to go faster or …

1___ 2___ 3___ 4___ 5___ 6___ 7___ 8___ 9___ 10___

**Friction in Everyday Life**

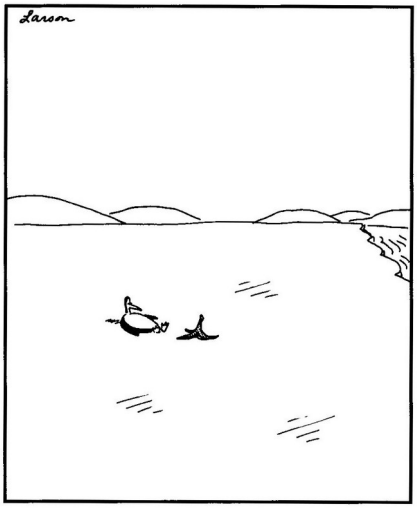
# Forced to Move

We use friction everyday. Sometimes we need friction, and at other times we try to reduce it.

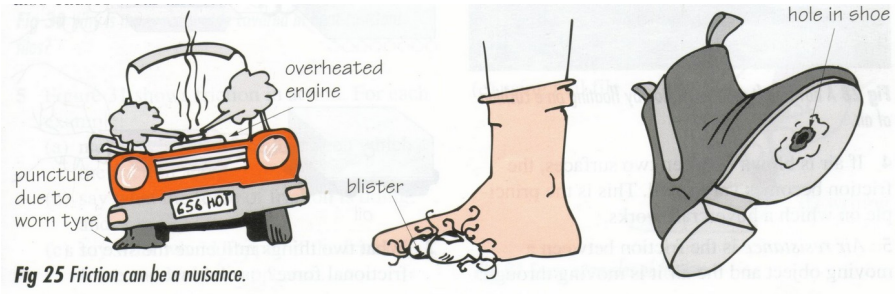
*Need friction*? We need it for example when:

- we walk. We use the friction between our feet ground. Imagine trying to walk if there was no f This would be like walking on ice!
- we drive a car. You could not stop a car friction. You could not start it moving either wheels would just spin without the car movin everybody knows what happens if you go t around a corner or if the road is slippery.
- we use knots or nails. Friction stops knots from undone, and holds nails in place.

The list goes on and on.




However, friction can also be a hassle. It opposes motio Friction also produces heat. This can be useful when you rub a match on the side of a box, but it can also cause an engine to overheat. Friction also causes wear and tear.




**Friction can be a nuisance.**

## Reducing Friction

There are five different ways of reducing friction.

1. A **rolling** object meets with less friction than a sliding one. (eg: moving a car on wheels, Egyptions moving huge blocks on rolling logs, ball bearings).
2. **Lubricants** (LOO-bri-cants) such as oil and grease are often used to reduce friction between moving parts to reduce wear and tear and make them slide more easily (eg: in machines, on the bearings in bicycle and skate board wheels, between bones in our joints).
3. Surfaces in contact can be **polished** so that they slide over one another more easily. (eg: hulls in racing yachts).
4. If **air is blown between two surfaces**, the friction becomes very small. This is the principle on which a hovercraft works.
5. Air resistance is the friction between a moving object and the air it is moving through, and can be reduced by giving the object (eg: car, plane) a more streamlined shape. **Streamlining** is also important for objects moving through water (eg: surfboards, speedboats and fast-swimming fish).

## *Questions*

1. Give two examples of how we need friction, and of how friction can be a nuisance.

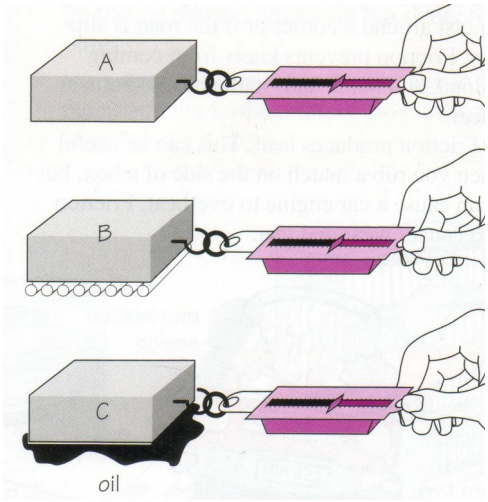
2. Use your knowledge of friction to briefly explain the following:
   a) Gymnasts put rosin on their hands before competing.

b) Surfers wax their surfboards.

c) A car uses more petrol when fitted with a roof rack.

d) When you drive your car down a long steep slope the brakes become quite hot.

3. Look at the adjacent diagrams. Copy and complete the sentences below using the correct words.
   a) Diagram A shows _____ (rolling/sliding) friction.
   b) The friction in B is _____ (greater/less) than in A.
   c) Rolling friction is _____ (greater/less) than
   d) sliding friction.
   e) When the object slides there is _____ (more/less) resistance to movement than when it rolls.
   f) With lubrication (diagram C) you need _____ (more/less) force to move an object.
   g) Lubrication _____ (increases/decreases) friction.

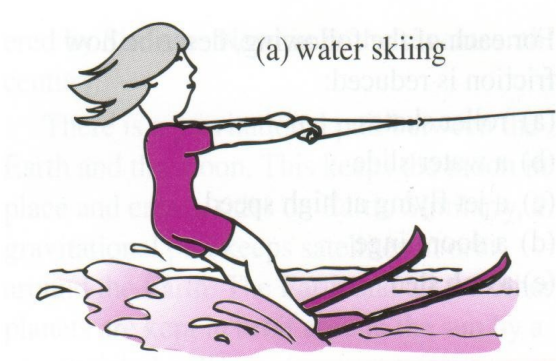


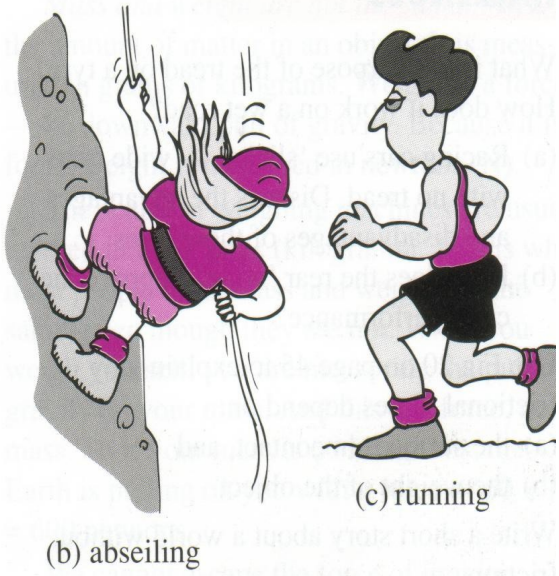

4. Link each of the following with how friction is reduced.

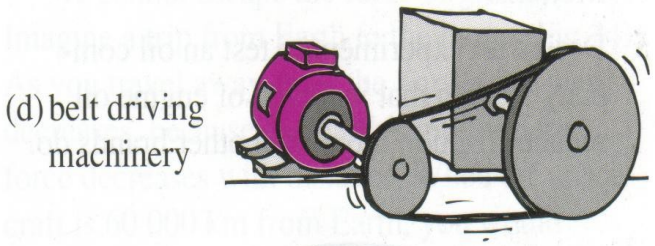
| | |
|---|---|
| roller skates | lubrication |
| concorde (fast airplane) | polishing |
| door hinge | ball-bearings |
| bob sled | streamlining |

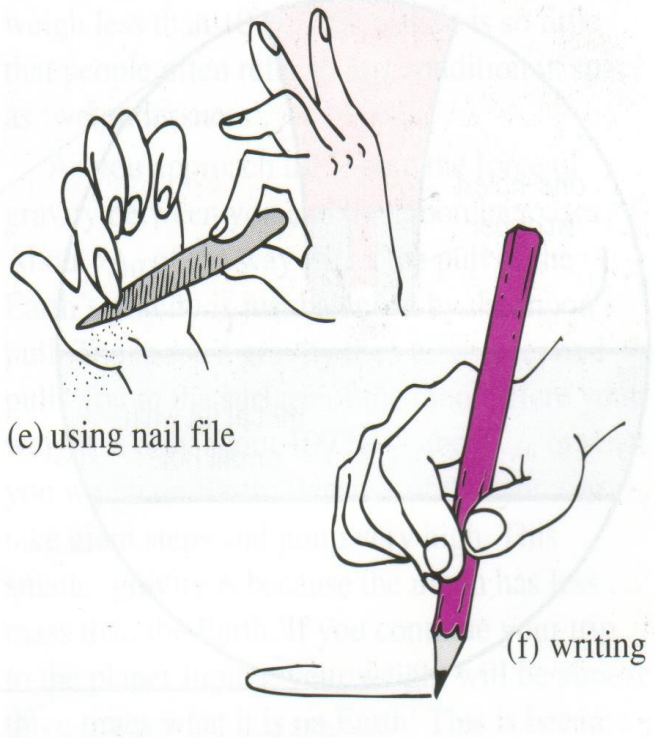
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5. The figures below show friction in action. For each example:
   a) name the two surfaces between which the friction acts
   b) say what the force of friction is doing, and
   c) say what would happen if the friction force suddenly disappeared.



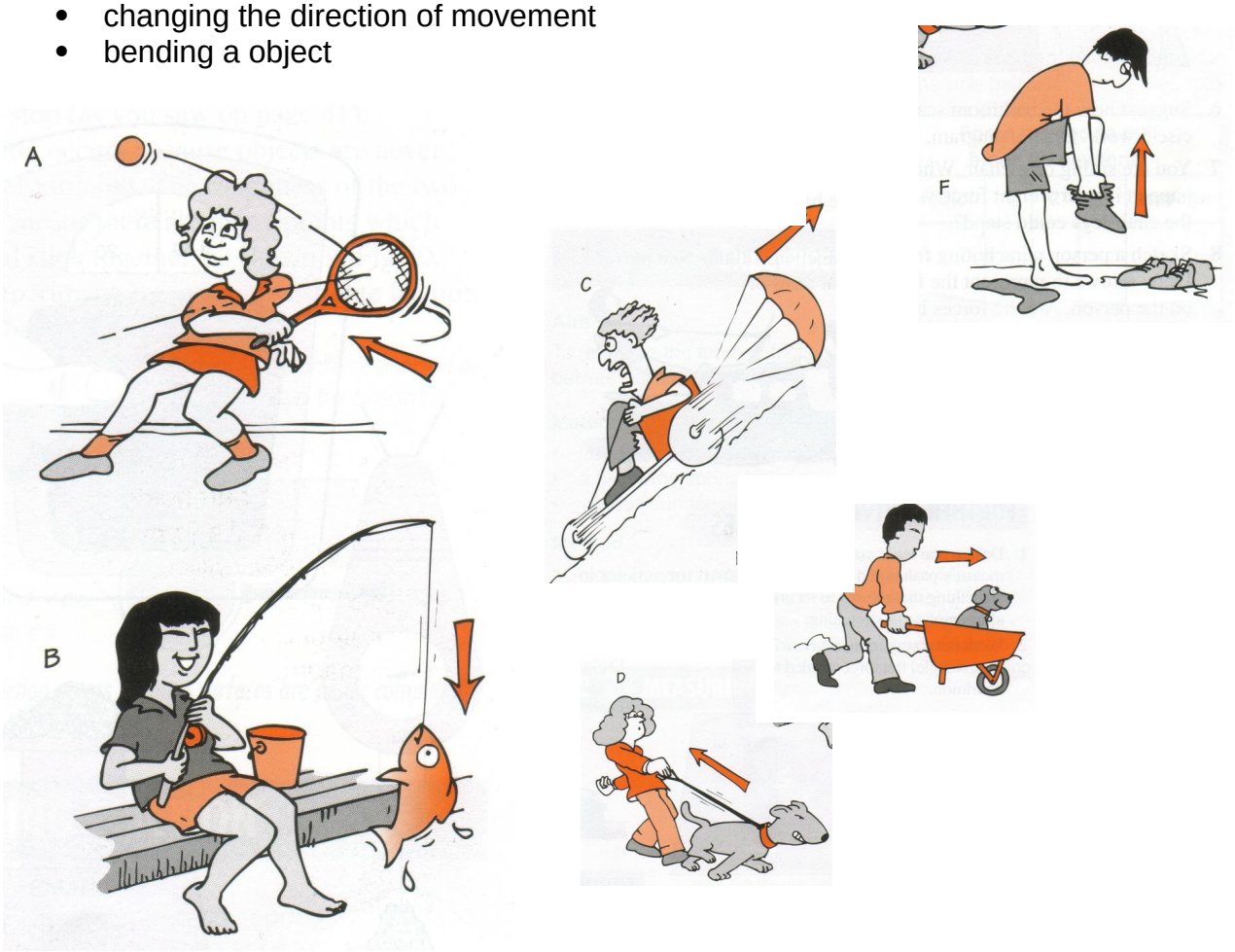

| |
|---|
| The |
| The |
| If t |
| The |
| The |
| If the friction force suddenly disappeared then |
| RUNNING |
| The two surfaces between which the friction acts are |
| The force of friction is.. |
| If the friction force suddenly disappeared then |
| USING A NAIL FILE |
| The two surfaces between which the friction acts are |
| The force of friction is.. |
| If the friction force suddenly disappeared then |

# Forced to move

**FORCES IN ACTION**

The diagrams below show some forces in action. The forces are shown with arrows. For each picture:

a) Name the object that the force acts on; for example, in A the force acts on the ball.
b) Choose from the list below what the force is doing in each case.
   - starting an object moving
   - stopping an object that is moving
   - changing the direction of movement
   - bending a object




**a.**

**b.**

**c.**

**d.**

**e.**

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**Questions**

1. Sketch a person accelerating on a skateboard, and draw arrows representing the forces acting.

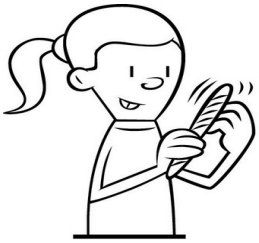
2. Imagine you are falling down a cliff. What are the forces acting on you? Are they balanced?

# Forced to move

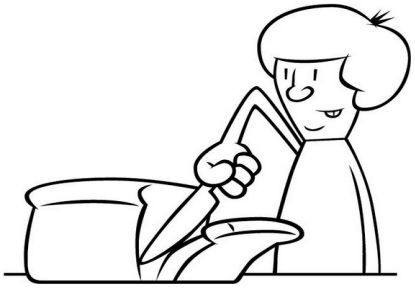
## Friction

Have you ever tried to help someone start a car by pushing it? It is very hard to get the car rolling right at the start. If you can get the car to move it seems to be easier to push once it has started to move. The car tyres have a force working against them. This force is trying to stop the car going forward or is said to be resisting the movement. This force is called *friction*.

**Friction is all around us. It is because of friction that erasers work for us. There is friction**







**working against us when we ride a bike or skateboard, roller blade, run, play tennis, cut our food and use a nail file to shape our nails. These are just a few examples of friction at work. There are several different types of friction:**

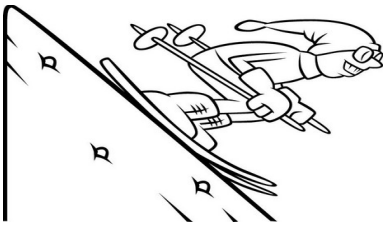
- Static friction is the force that stops things from starting to move: think of trying to push a car.
- Sliding friction happens between moving surfaces such as in downhill skiing.
- Rolling friction happens when a curved surface rolls over another surface. This happens when the wheel of a bike turns in one direction and the bike moves in the other direction.
- Fluid friction happens between fluids or between fluids and solids. Fluid friction is less than friction between solids. Fluids are often used to help movement of two solids by reducing the friction between the surfaces. The axle on a bike has oil on it to help the metal axle turn more easily inside the casing.

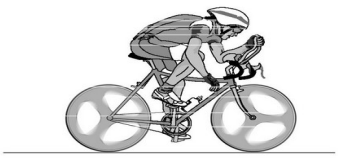
**Activity**

**Here are some pictures of different types of friction.**

**1** Label each picture static, sliding, rolling or fluid friction in the space.

In the space provided write down why you think the example is static friction, sliding friction, rolling friction or fluid friction.





I think this is _____ friction

because _____

_____

I think this is _____ friction

because _____

_____

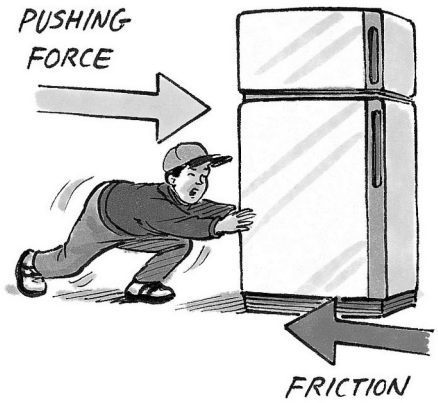
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I think this is _____ friction

because _____

_____




I think this is _____ friction

because _____

_____

**2** Think about this. Why is it sometimes very difficult to stop a car on a wet day?

**Write down your answer in the space provided.**

_____

_____

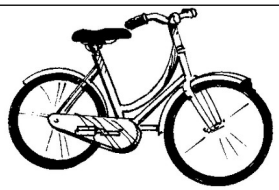
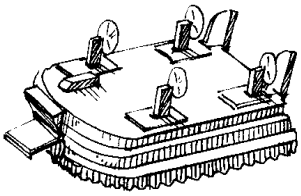
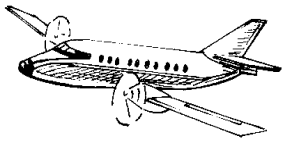
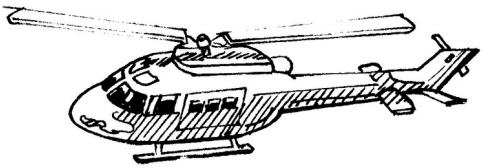
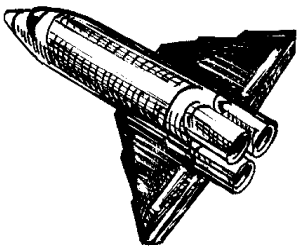
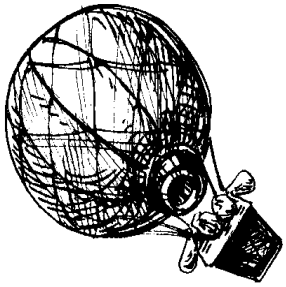
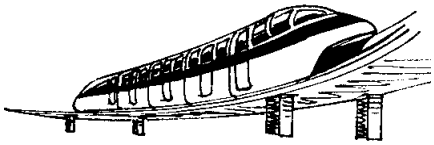
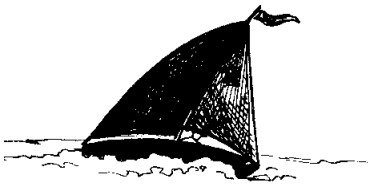
_____

_____

_____

# Forced to move

Look at the following examples of ways in which we travel. List the forces that are acting in each case (gravity, friction, pushes, pulls, twists, etc.).

| | |
|---|---|
|  |  |
| Bicycle | Hovercraft |
|  |  |
| Truck | Plane |
|  |  |
| Helicopter | Rocket |
|  |  |
| Hot air balloon | Monorail |
|  |  |
| Punt | Yacht |

The punt and the yacht both float on water and move forward. Describe the forces which act on these vessels when they are moving. How are they the same? How are they different?

# Forced to move

-

1. A. Link the following examples to the appropriate forces

| | |
|---|---|
| A compass | Friction |
| Bungie jumping | Magnetic |
| Sandpaper | Gravitational |
| Sticky tape | Adhesion |
| Water droplet | Cohesion |
| Charged hair | Electrical |
| Screwdriver | Mechanical |

B. From the above select contact and non contact forces

2. Complete the sentence
The effect of a force on a body is to change its_____ or its_____.

3. Consider a car. Describe two ways in which this machine uses the forces of friction and two ways in which the force is reduced.

4. Answer the questions below from the following table:

| **Planet** | **Weight of brick (N)** | **Mass of Brick (kg)** |
|---|---|---|
| Earth | 20 | 2 |
| Jupiter | 50 | 2 |
| Saturn | 25 | 2 |
| Pluto | 3 | 2 |

a) On which planet was the weight of the brick the greatest?
b) On which planet was the weight of the brick the least?
c) Therefore which is the most massive planet?
d) On which planet was the mass of the brick the greatest?

# Forced to move

5. An object is balanced on its edge such that part of it hangs over the edge of a table. If the object is stationary, what must be true about the system of forces acting on it? Give examples of the forces acting.

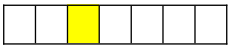
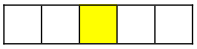
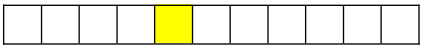
6. Name the three types of non-contact forces?

# Forced to move

QUESTION: *Why don't tourists go to the moon?*

ANSWER*: Because it doesn't _ _ _ _ _ _ _ people like the earth.*

*To find the answers, first unscramble the seven words. There are clues besides each one to help you. Write the unscrambled words in the boxes, the shaded letters spell out the missing word in the answer to the question.*

1. *This force helps keep us on the earth*
   *T A I G V Y R*
   
2. *This force slows down moving objects*
   *I R N F O C T I*
   
3. *This is a type of contact force*
   *W S N I T G T I*
   
4. *This thing causes changes*
   *E R F C O*
   
5. *This kind of force must touch an object before it can make something happen*
   *N O T C T C A*
   
6. *These provide the force needed to move the body*
   *C E M U L S S*
   
7. *This is a type of non-contact force*
   *R E Y C E C T L I T I*
   

# Forced to move

## Measuring weight and mass

If you put a one-kilogram bag of sugar on a set of scales, it is the gravitational pull on the sugar which forces the scale pan downwards-the scales respond to weight.
Some sets of scales are marked in newtons but most give a reading in mass units such as kilograms or grams (there are 1000 grams in each kilogram). Readings in mass units present no problem to manufacturers of scales because an object of large mass also has a large weight on Earth.
As for the one-kilogram bag of sugar: it has a moss of 1 kilogram and a weight on Earth of 9.8 newtons, but try telling them that at the supermarket!

**Questions**

1. In what units are forces measured?

2. In what units are masses measured?

3. What is the common name for gravitational force?

4. Why does an astronaut weigh less on the Moon than the Earth? How is his mass affected if he moves from the Earth to the Moon?

5. How many grams are there in 1 kilogram?

6. Why is it wrong to say that a bag of sugar 'weighs' 1 kilogram? What is the weight of the sugar on Earth?

8. Use the conversion 1 kg wt = 9.8 newtons to complete the table.

| MASS | WEIGHT |
|---|---|
| 5 kg | |
| 48 kg | |
| 0.2 kg | |
| 0.85 kg | |
| | 30 N |
| | 182 N |
| | 7 N |
| 250 g | |
| | |

9. If a litre of water has a mass of 1 kg, how much water is needed top produce a force of gravity of
   1. 20 N  2. 150 N  3. 1000 N  4. 4 N ?

# Forced to move

## Gravity

You can jump in the air but you cannot stay there. You are pulled down by the force of **gravity**. When you drop a glass or something falls, it is because gravity pulls the object towards the Earth. This gravitational force gives objects their weight.

Gravity is the pull of the enormous mass of the Earth on our bodies and any other objects near the Earth's surface. It is a non-contact force since it is able to act over a distance, without the objects touching each other.

There is a small gravitational force that pulls any two objects together. The size of this force depends on:

- the masses of the objects, and
- how far apart they are.

Usually this force is very small, but when one of the objects is the planet Eart, the force is far from small. This law of gravitation was discovered by Sir Isaac Newton in the seventeenth century.

There is a gravitational pull between the Earth and the moon. This keeps the moon in place and causes tides on Earth. Similarly, a gravitational pull keeps satellites in orbit around the Earth. The Earth and other planets are kept in orbit around the sun by a gravitational pull.

### *Questions*

1. a) What two things control the size of the gravitational force between two objects?

   a) Why then is the force of gravity between two apples on a table too small to notice?

   b) Why is the force of gravity between the Earth and the star Rigel too small to notice?

   c) Why is the force of gravity between the sun and its planets strong enough to hold them in orbit?

2. Suggest why the force of gravity is less on the moon?

# Forced to move

Speed

Working out speeds!

**Average speed = distance/time or average speed is the distance travelled in the time it took to go that distance.**
**Average speed has the symbol *s*. The symbol for distance is *d* and the symbol for time is *t*.**

**So average speed = $\frac{\text{distance}}{\text{time}}$ or $s = \frac{d}{t}$**

**Let's work out some speeds.**
**Jane rode her bike 40 kilometres in 2 hours. We want to know speed or *s*.**
**We know distance or *d* = 40 km. We know time or *t* = 2 h.**

**If $s = \frac{d}{t}$ then $s = \frac{40}{2}$ km/h**
**= 20 km/h**

**Always set out your calculations like this and you will find it easier to understand the facts that you have been given.**

**Activity**

**1** The distance from Coolangatta to Caloundra is 180 kilometres and it takes Marie 2 hours to make the trip. What is the average speed of Marie's car in km/h? Write your calculation in the space below.

Use $s = \frac{d}{t}$ distance = 180 km time = 2 h

_____

_____

**2** Joel walked 18 kilometres in 3 hours. What was his average speed in km/h? Set your calculations out like the example above.

_____

_____

**3** Olympic runners can run 100 metres in about 9 seconds. What is their average speed in m/s?

_____

_____

**A challenge!**

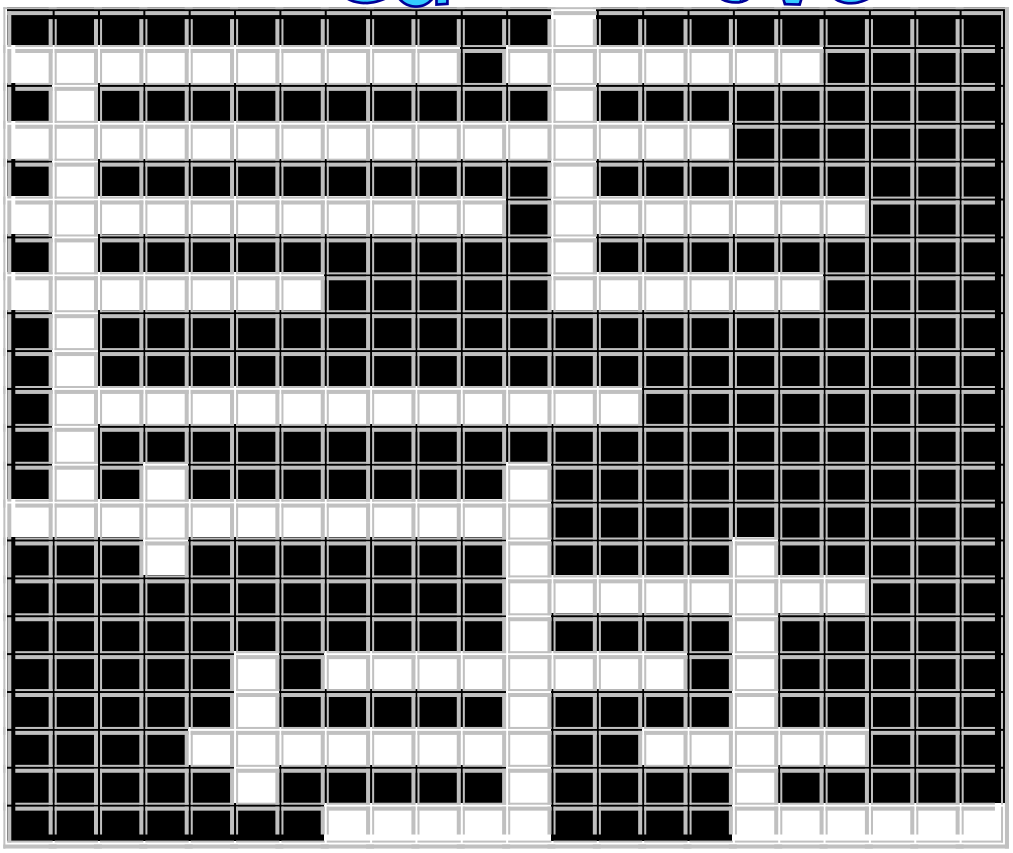
**4** If a Grand Prix car travels at 220 km/h, how long would it take it to travel 100 kilometres? Can you work this out? If you like you can work with a friend to solve this challenge.

_____

_____

**Forces Crossword**

# Forced to move



Down:

| | |
|---|---|
| 1 | THIS FORCE SLOWS US DOWN |
| 3 | A FORCE ON A CHARGED ROD OR COMB |
| 11 | THE TOTAL AMOUNT OF FORCES |
| 12 | FORCES ARE THIS WHEN AN OBJECT MOVES |
| 14 | THE SI UNIT FOR WEIGHT |
| 16 | ANOTHER NAME FOR FRICTION |